REDUCTION TO RATIONAL SINGULARITIES ASCENDS ALONG SEPARABLE FINITE LOCAL EXTENSIONS OF NORMAL SURFACE SINGULARITIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the production of a nonzero determinant from the nondegenerate trace pairing, the reduction to a normal modification finite over a modification of the source via an admissible blowup, and the comparison of the first cohomology groups through the trace map together with the bound on the remaining torsion.

Editorial note. The label lemma-go-up-separable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-go-up-separable –start 2138 –end 2202 –local-id 28084 returned RESULT=OK (L28084 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Nondegeneracy of the trace pairing for a finite separable extension. Fields, the lemma that the trace pairing of a finite separable field extension is nondegenerate. Supplies the nonvanishing of the determinant of the matrix of traces of a basis, the element used to bound torsion below.
- Finiteness of a strict transform after an admissible blowup. More on Flatness, the lemma producing an admissible blowup after which the strict transform becomes finite. Supplies the reduction to a situation where the modification of the target is finite over a modification of the source, both normal.
- Boundedness of the torsion in the first cohomology. Same chapter, the lemma bounding the torsion of the first cohomology by a fixed element. Supplies the final ingredient completing the induction of the vanishing.

Source reference. Original equation 0AXK: sources/stacks-resolve/pinned-original-source.tex, source lines 1965–1965. Prerequisite pointer invoked at proof line(s) 2201; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BIL: sources/stacks-chow/tags.txt, source lines 14494–14494. Prerequisite pointer invoked at proof line(s) 2151; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B49: sources/stacks-chow/tags.txt, source lines 13992–13992. Prerequisite pointer invoked at proof line(s) 2159; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AXJ: sources/stacks-resolve/pinned-original-source.tex, source lines 1947–1947. Prerequisite pointer invoked at proof line(s) 2200; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $A \to B$ be a finite injective local ring map of local normal Nagata domains of dimension 2. Assume that the induced extension of fraction fields is separable. If reduction to rational singularities is possible for A then it is possible for B.*

Proof. Let n be the degree of the fraction field extension L/K. Let $\mathrm{Trace}_{L/K} : L \to K$ be the trace. Since the extension is finite separable the trace pairing $(h, g) \mapsto \mathrm{Trace}_{L/K}(fg)$ is a nondegenerate bilinear form on L over K. See Fields, Lemma 0BIL. Pick $b_1, \ldots, b_n \in B$ which form a basis of L over K. By the above $d = \det(\mathrm{Trace}_{L/K}(b_i b_j)) \in A$ is nonzero.

Let $Y \to \mathrm{Spec}(B)$ be a modification with Y normal. We can find a U-admissible blowup X' of $\mathrm{Spec}(A)$ such that the strict transform Y' of Y is finite over X', see More on Flatness, Lemma 0B49. Picture

$$\begin{array}{ccccc} Y' & \longrightarrow & Y & \longrightarrow & \mathrm{Spec}(B) \\ \downarrow & & & & \downarrow \\ X' & & \longrightarrow & & \mathrm{Spec}(A) \end{array}$$

After replacing X' and Y' by their normalizations we may assume that X' and Y' are normal modifications of $\mathrm{Spec}(A)$ and $\mathrm{Spec}(B)$. In this way we reduce to the case where there exists a commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{g} & \mathrm{Spec}(B) \\ {\scriptstyle\pi}\downarrow & & \downarrow \\ X & \xrightarrow{f} & \mathrm{Spec}(A) \end{array}$$

with X and Y normal modifications of $\mathrm{Spec}(A)$ and $\mathrm{Spec}(B)$ and π finite.

The trace map on L over K extends to a map of $\mathcal{O}_X$-modules $\mathrm{Trace} : \pi_*\mathcal{O}_Y \to \mathcal{O}_X$. Consider the map

$$\Phi : \pi_*\mathcal{O}_Y \longrightarrow \mathcal{O}_X^{\oplus n}, \quad s \longmapsto (\mathrm{Trace}(b_1 s), \ldots, \mathrm{Trace}(b_n s))$$

This map is injective (because it is injective in the generic point) and there is a map

$$\mathcal{O}_X^{\oplus n} \longrightarrow \pi_*\mathcal{O}_Y, \quad (s_1, \ldots, s_n) \longmapsto \sum b_i s_i$$

whose composition with Φ has matrix $\mathrm{Trace}(b_i b_j)$. Hence the cokernel of Φ is annihilated by d. Thus we see that we have an exact sequence

$$H^0(X, \mathrm{Coker}(\Phi)) \to H^1(Y, \mathcal{O}_Y) \to H^1(X, \mathcal{O}_X)^{\oplus n}$$

Since the right hand side is bounded by assumption, it suffices to show that the d-torsion in $H^1(Y, \mathcal{O}_Y)$ is bounded. This is the content of Lemma 0AXJ and (0AXK). □